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# Identification of a Self-Optimizing Control Structure from Normal Operating Data

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***Abstract*— In building systems, it is necessary to maintain comfort conditions while minimizing operating costs. One way to do this is to use a real-time optimizer (RTO) to adjust the setpoints of the controlled variables. Another way is to use a self-optimizing control (SOC) structure to find one or more new variables that, when controlled to the appropriate constant setpoints, drive the operating cost of the system to, or close to, its optimal point. A requirement of SOC is that an optimal operating point be used in the calculation of the self-optimizing variable(s) and the identification of its parameters. In this work, we propose a formulation of SOC that allows for the use of non-optimal data. The proposed method makes use of normal operating data to identify the parameters used to calculate the self-optimizing variables. Simulation of an HVAC system shows that the performance of the proposed method is similar to that of SOC based on optimal data, and also to the performance of an RTO alternative based on extremum-seeking control (ESC).**

## I. Introduction

Systems in buildings are primarily designed to maintain comfort conditions for occupants. The complexity and redundancy present in most building systems make it possible to satisfy comfort conditions in different ways. One example of this is when cold air is supplied to a room to maintain a temperature setpoint. In this case, the setpoint could be maintained through different combinations of supply air temperature and flow rate, e.g., room temperatures could be lowered by either increasing air flow or reducing the temperature of the supply air. The cost of air flow is governed by the fan and the cost of lowering the air temperature is dominated in most cooling applications by compressor energy. A trade-off therefore exists that can be exploited in order to minimize total costs whilst still satisfying comfort.

The trade-offs inherent in building systems can be managed by applying optimization methods that make adjustments to system operation. For the air flow example, the temperature of the air delivered to the room is usually controlled to a fixed setpoint. An optimization method could be used to adjust this setpoint to minimize the total cost of the fans and chiller plant. This situation is common in complex systems where a network of controllers operate to maintain setpoints and an optimization method is used to adjust the setpoints based on a desired cost function. Standard terminology for this optimization functionality is "real-time optimization layer", or RTO layer.

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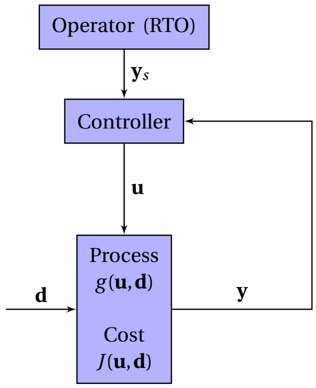
Self-optimizing control (SOC) was developed by recognizing that the RTO layer is only needed because the control layer that comprises the feedback controllers essentially controls the wrong variables from a holistic control and optimization perspective [1]. Conventional feedback control structures are usually designed for regulation to constant setpoints, which satisfies control objectives but leads to optimization cost function measures varying with system operating points and disturbances. The purpose of SOC is to design the control structure so that regulation at constant setpoints maintains not only control objectives but also optimization targets.

The concept of SOC is illustrated in Figure 1 which shows the conventional RTO approach (Fig. 1a) and the SOC approach (Fig. 1b). The conventional approach has a control layer and an RTO layer, whereas the SOC approach creates a new control structure that contains a set of control variables that are subject to regulatory control at constant setpoints. The control structure is designed such that both control and optimization objectives are satisfied simultaneously through the regulatory action of the feedback controller. The key to the approach is the creation of a new set of control variables $\mathbf{c}$. These variables are derived from a linear (or non-linear) mapping of available measurements whereby the mapping is usually obtained from operational data obtained at optimal operating points. For example, this would require gathering data from the conventional layered strategy and recording data values at the times when the RTO layer achieved optimality.

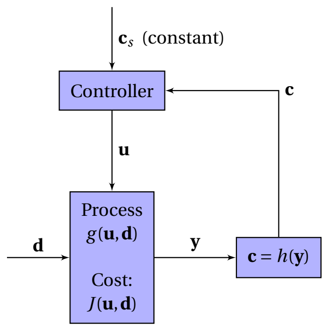
The method proposed in this work enables the mapping function to be obtained without the need for optimal data. Instead, the proposed method can work by using normal operating data across a range of conditions. Section II provides a description of SOC; Section III presents a derivation of the self-optimizing (SO) control variable using non-optimal data. Sections IV and V show the description of the VAV system used to test the proposed SOC method, as well as the simulations performed and the obtained results. Finally, conclusions are given in Section VI.

## II. Self-Optimizing Control

SOC is a control scheme that uses a combination of measured variables to create new variables that, when kept at the appropriate constant setpoint, keeps the controlled system at optimal or near-optimal operating conditions regardless of disturbances in the system [1]. Figure 1b shows a representation of how SOC works; the controlled process takes inputs $\mathbf{u}$ and disturbances $\mathbf{d}$, and produces




(a) Conventional approach.




(b) Self-optimizing control approach.

Fig. 1: Comparison of the conventional control structure and the SOC structure.

variables $\mathbf{y}$ as

$$\mathbf{y} = g(\mathbf{u},\mathbf{d}) \tag{1}$$

The number of inputs, disturbances and measured variables are $n_u$, $n_d$ and $n_y$, respectively; which are also the sizes of the vectors $\mathbf{u}$, $\mathbf{d}$ and $\mathbf{y}$. The process has an operating cost given by a function $J(\mathbf{u},\mathbf{d})$. The measurements $\mathbf{y}$ are used to calculate a variable $\mathbf{c}$, which is fed to a controller that keeps $\mathbf{c}$ at a setpoint $\mathbf{c}_s$. In order to do this, the controller will calculate values of $\mathbf{u}$ that, when applied to the process will make the cost $J(\mathbf{u},\mathbf{d})$ move to, or very close to, its optimal value $J^{opt}(\mathbf{u}^{opt},\mathbf{d})$.

Several approaches have been developed to calculate $\mathbf{c}$; some are model-based ([1], [2]) and others are data-driven ([3], [4], [5], [6], [7], [8]). A survey of SOC methods is shown in the work of Jäschke et al. [9]. The work in this project is inspired by the data-driven method from Jäschke and Skogestad [6].

The variable $\mathbf{c}$, which we will refer to as the self-optimizing (SO) variable or SOC variable, can be thought of as the gradient of the cost function $J(\mathbf{u},\mathbf{d})$ with respect to $\mathbf{u}$. In a data-driven approach, $\mathbf{c}$ can be calculated in general as a linear combination of the measured variables as

$$\mathbf{c} = \mathbf{h}_0 + \mathbf{H}\Delta\mathbf{y} \tag{2}$$

where $\mathbf{h}_0$ is an offset vector of size $n_u$ and $\mathbf{H}$ is an $n_u \times n_y$ matrix. Furthermore, $\Delta\mathbf{y} = \mathbf{y} - \mathbf{y}^*$, where $\mathbf{y}^*$ is a reference, or nominal, point from which the parameters $\mathbf{h}_0$ and $\mathbf{H}$ are calculated. In the existing literature, the reference point $\mathbf{y}^*$ is selected as the optimal operating condition of the plant for a given disturbance. In this work we extend the definition of the reference point to be any operating condition where variables are within their physical bounds, i.e., none of the variables are saturated.

We use the terms reference point and nominal point interchangeably. Also, we use different superscripts to differentiate the nominal operating conditions. In general, we refer to any nominal point with the superscript $*$; for example $\mathbf{y}^*$. For the special case of an optimal nominal point we use the superscript $opt$; for example $\mathbf{y}^{opt}$. For any non-optimal reference point we will use the superscript $r$; for example $\mathbf{y}^r$.

### A. Self-optimizing control from optimal data

For a given disturbance $\mathbf{d}^*$, the plant has an optimal cost $J^{opt}(\mathbf{u}^{opt},\mathbf{d}^*)$ and an optimal input $\mathbf{u}^{opt}$. The optimal input can be found with an optimizer, such as the RTO layer in Figure 1a, an example being an Extremum Seeking Controller (ESC) [10]. The optimal output of the process is

$$\mathbf{y}^{opt} = g(\mathbf{u}^{opt},\mathbf{d}^*) \tag{3}$$

which will be our reference point in the design of the SOC controller; this is $\mathbf{y}^* = \mathbf{y}^{opt}$. Moreover, since the variable $\mathbf{c}$ is the gradient of the cost function with respect to the process input, this value should be zero at the optimal operating condition; thus, when $\mathbf{y} = \mathbf{y}^{opt}$, $\mathbf{c} = 0$ and, from Equation 2, $\mathbf{h}_0 = \mathbf{0}$. This means that for an optimal reference point, the SO variable is calculated as

$$\mathbf{c} = \mathbf{H}\Delta\mathbf{y} \tag{4}$$

This is the expression for $\mathbf{c}$ found in [3], [4], [5], [6], [7], [8].

### B. Self-optimizing control from non-optimal data

If optimal data are not available, we can pick any operating condition to calculate the SO variable. A non-optimal input $\mathbf{u}^r$ will produce the output

$$\mathbf{y}^r = g(\mathbf{u}^r,\mathbf{d}^*) \tag{5}$$

We use this output as our reference point $\mathbf{y}^* = \mathbf{y}^r$. The only requirement is that the input $\mathbf{u}^r$ is not at a constraint. The SO variable is then calculated as Equation 2, with $\Delta\mathbf{y} = \mathbf{y} - \mathbf{y}^r$. This is a more general expression and is the one we use in this work.

## III. Calculation of the Self-Optimizing Variable

Using a second-order Taylor approximation, the cost function $J(\mathbf{u},\mathbf{d})$ can be approximated around the disturbance $\mathbf{d}^*$ and the general reference input $\mathbf{u}^*$ as

$$\Delta J = \begin{bmatrix} \mathbf{J}_u^* \\ \mathbf{J}_d^* \end{bmatrix}^T \begin{bmatrix} \Delta\mathbf{u} \\ \Delta\mathbf{d} \end{bmatrix} + \frac{1}{2} \begin{bmatrix} \Delta\mathbf{u} \\ \Delta\mathbf{d} \end{bmatrix}^T \begin{bmatrix} \mathbf{J}_{uu}^* & \mathbf{J}_{ud}^* \\ \mathbf{J}_{du}^* & \mathbf{J}_{dd}^* \end{bmatrix} \begin{bmatrix} \Delta\mathbf{u} \\ \Delta\mathbf{d} \end{bmatrix} \tag{6}$$

where $\Delta\mathbf{u} = \mathbf{u} - \mathbf{u}^*$, $\Delta\mathbf{d} = \mathbf{d} - \mathbf{d}^*$, $\Delta J = J(\mathbf{u},\mathbf{d}) - J(\mathbf{u}^*,\mathbf{d}^*)$, $\mathbf{J}_u^* = \frac{\partial J}{\partial \mathbf{u}}\Big|_{\mathbf{u}=\mathbf{u}^*,\mathbf{d}=\mathbf{d}^*}$, $\mathbf{J}_d^* = \frac{\partial J}{\partial \mathbf{d}}\Big|_{\mathbf{u}=\mathbf{u}^*,\mathbf{d}=\mathbf{d}^*}$, $\mathbf{J}_{uu}^* = \frac{\partial^2 J}{\partial \mathbf{u}\partial \mathbf{u}^T}\Big|_{\mathbf{u}=\mathbf{u}^*,\mathbf{d}=\mathbf{d}^*}$,

$\mathbf{J}_{ud}^* = \frac{\partial^2 J}{\partial \mathbf{u} \partial \mathbf{d}^T}\Big|_{\mathbf{u}=\mathbf{u}^*, \mathbf{d}=\mathbf{d}^*}$, $\mathbf{J}_{du}^* = \frac{\partial^2 J}{\partial \mathbf{d} \partial \mathbf{u}^T}\Big|_{\mathbf{u}=\mathbf{u}^*, \mathbf{d}=\mathbf{d}^*}$, $\mathbf{J}_{dd}^* = \frac{\partial^2 J}{\partial \mathbf{d} \partial \mathbf{d}^T}\Big|_{\mathbf{u}=\mathbf{u}^*, \mathbf{d}=\mathbf{d}^*}$. Furthermore, the gradient of the cost function with respect to $\mathbf{u}$, $\mathbf{J}_u$, can be approximated as

$$\mathbf{J}_u = \mathbf{J}_u^* + \begin{bmatrix} \mathbf{J}_{uu}^* & \mathbf{J}_{ud}^* \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{d} \end{bmatrix} \tag{7}$$

In general, the disturbance $\Delta \mathbf{d}$ cannot be measured directly; it can, however, be calculated from the process measurements. From Equation 1, the measurements $\mathbf{y}$ can be approximated around the reference point as

$$\Delta \mathbf{y} = \begin{bmatrix} \mathbf{G}_u^* & \mathbf{G}_d^* \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{d} \end{bmatrix} \tag{8}$$

where $\Delta \mathbf{y} = \mathbf{y} - \mathbf{y}^*$ is the deviation from the reference point, and

$$\mathbf{G}_u^* = \frac{\partial \mathbf{y}}{\partial \mathbf{u}^T}\Big|_{\mathbf{u}=\mathbf{u}^*, \mathbf{d}=\mathbf{d}^*} \tag{9}$$

and

$$\mathbf{G}_d^* = \frac{\partial \mathbf{y}}{\partial \mathbf{d}^T}\Big|_{\mathbf{u}=\mathbf{u}^*, \mathbf{d}=\mathbf{d}^*} \tag{10}$$

are the static gains of the measurements $\mathbf{y}$ with respect to the inputs $\mathbf{u}$ and disturbance $\mathbf{d}$, respectively. Making

$$\mathbf{G}_y = \begin{bmatrix} \mathbf{G}_u^* & \mathbf{G}_d^* \end{bmatrix} \tag{11}$$

we get

$$\begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{d} \end{bmatrix} = \mathbf{G}_y^+ \Delta \mathbf{y} \tag{12}$$

The term $\mathbf{G}_y^+$ is the pseudo-inverse of $\mathbf{G}_y$. In order for $\mathbf{G}_y$ to be invertible, the number of measured variables should be greater than or equal to the number of inputs plus the number of disturbances, this is, $n_y \geq n_u + n_d$. Substitution of Equation 12 in $\mathbf{J}_u$ leads to

$$\mathbf{J}_u = \mathbf{J}_u^* + \begin{bmatrix} \mathbf{J}_{uu}^* & \mathbf{J}_{ud}^* \end{bmatrix} \mathbf{G}_y^+ \Delta \mathbf{y} \tag{13}$$

If we look at Equation 13, it has the same form as Equation 2; thus, if we make $\mathbf{c} = \mathbf{J}_u$ we have

$$\mathbf{h}_0 = \mathbf{J}_u^* \tag{14}$$

$$\mathbf{H} = \begin{bmatrix} \mathbf{J}_{uu}^* & \mathbf{J}_{ud}^* \end{bmatrix} \mathbf{G}_y^+ \tag{15}$$

For the special case of having optimal data, $\mathbf{h}_0 = \mathbf{J}_u^{opt} = 0$, which makes $\mathbf{c} = \mathbf{H}\Delta \mathbf{y}$.

Controlling $\mathbf{c}$ to zero will lead to the optimal operation of the plant. However, at this point it is not yet possible to compute $\mathbf{c}$ since we do not know what $\mathbf{h}_0$ is and, as seen in Equation 15, $\mathbf{H}$ depends on derivatives of the cost $J$ and outputs $\mathbf{y}$ with respect to the disturbances $\mathbf{d}$. A calculation of these values is shown next.

### A. Calculation of the offset vector and combination matrix

The calculation of $\mathbf{H}$ and $\mathbf{h}_0$ is the most important step in the design of the SOC structure. The calculate $\mathbf{H}$, we use the method by Jäschke and Skogestad [6], where they they show

$$\mathbf{H} = (\mathbf{G}_u^*)^T \mathbf{J}_{yy}^* \tag{16}$$

where $\mathbf{G}_u^*$ is defined in Equation 9, and

$$\mathbf{J}_{yy}^* = \frac{\partial^2 J}{\partial \mathbf{y} \partial \mathbf{y}^T}\Big|_{\mathbf{y}=\mathbf{y}^*} \tag{17}$$

is the Hessian of the cost function $J$ with respect to the measured variables $\mathbf{y}$ at the reference point $\mathbf{y}^*$. Appendix A shows a derivation of these relationships.

The calculation of $\mathbf{h}_0$, one of the contributions of this work, is obtained as

$$\mathbf{h}_0 = (\mathbf{G}_u^*)^T \mathbf{J}_y^* \tag{18}$$

where

$$\mathbf{J}_y^* = \frac{\partial J}{\partial \mathbf{y}}\Big|_{\mathbf{y}=\mathbf{y}^*} \tag{19}$$

is the gradient of the cost $J$ with respect to the measurement $\mathbf{y}$ evaluated at the reference point $\mathbf{y}^*$. This result is also shown in Appendix A.

## IV. Control of a VAV System

The performance of proposed SOC method with normal operating data is tested on a variable air volume (VAV) system. We compare the performance of the method with those of SOC with optimal data, ESC, and standard control. By standard control we mean a structure where the controlled variables are kept at fixed setpoints regardless of changes in disturbances. We also use ESC to find the optimal operation of the system and use it as a benchmark for optimal performance. For simplification purposes we will refer to the SOC method with non-optimal reference data as SOC-NOR.

Figure 2 shows a general representation of the VAV system used in the simulations. The goal of the system is to keep the temperature in the zone $T_{zn}$ at a desired value $r_{zn}$, regardless of the outdoor temperature and load conditions. In order to achieve the zone temperature setpoint, the fan speed $f_s$ is manipulated with a controller $PI_{zn}$. The fan blows supply air at a temperature $T_{sa}$ that it is controlled by manipulating the compressor speed $k_s$. The expansion valve $v_p$ is controlled to maintain the superheat temperature $T_{sh}$ at a setpoint value $r_{sh}$. The controller block represents any of the four control structures used in the simulation. Each controller has different input variables; but, they all have the compressor speed $k_s$ as the common output. The following subsections show more details on how the controllers are implemented.

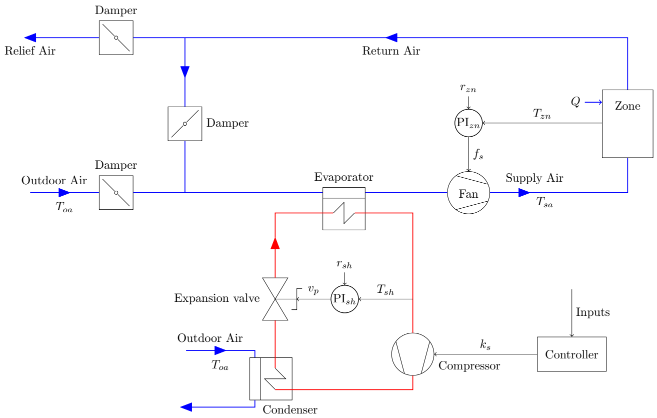


Fig. 2: VAV system.

### A. Standard control

In standard control, the supply air temperature setpoint $r_{sa}$ is fixed, and a PI controller is used to manipulate the compressor speed as

$$k_s = PI_{sa}(\bar{T}_{sa}) \tag{20}$$

where $\bar{T}_{sa} = r_{sa} - T_{sa}$ is the setpoint error of the supply air temperature. $PI_{sa}(\cdot)$ is a function that represents any discrete or continuous PI controller that drives the supply air temperature to its setpoint.

### B. Extremum seeking control

With ESC, the controller attempts to optimize operation of the VAV system by minimizing energy consumption while still satisfying zone loads. In this example, the optimization objective is defined as maximization of the coefficient of performance (COP) of the system. The COP is defined as the ratio of the heat removed in the evaporator divided by the combined power of the compressor and fan,

$$COP = \frac{Q_{evap}}{W_k + W_f} \tag{21}$$

It should be noted that the fan power associated with the condenser could also be included in the COP calculation, but, this is not considered in the present study.

Since ESC seeks to minimize an objective function, the negative value of the COP is used as input to the ESC. The ESC output is the supply air setpoint $r_{sa}$, that is

$$r_{sa} = ESC(-COP) \tag{22}$$

Here $ESC(\cdot)$ is the control sequence that finds the optimal value $r_{sa}^{opt}$ that drives the COP to its optimal value $COP^{opt}$. Once the optimal setpoint is found, the supply air temperature is driven to this value with the PI controller from Equation 20.

### C. Self-optimizing Control

The process variables selected to calculate the SO variable are the supply air temperature $T_{sa}$, the fan speed $f_s$ and valve position $v_p$. The SO variable is

$$c = h_0 + h_1\Delta T_{sa} + h_2\Delta f_s + h_3\Delta v_p \tag{23}$$

$$= h_0 + \mathbf{H}\Delta\mathbf{y} \tag{24}$$

with $\mathbf{H} = \begin{bmatrix} h_1 & h_2 & h_3 \end{bmatrix}$, $\Delta\mathbf{y} = \begin{bmatrix} \Delta T_{sa} & \Delta f_s & \Delta v_p \end{bmatrix}^T$, and $\Delta T_{sa} = T_{sa} - T_{sa}^*$, $\Delta f_s = f_s - f_s^*$ and $\Delta v_p = v_p - v_p^*$. The variables $T_{sa}^*$, $f_s^*$ and $v_p^*$ are the nominal conditions for the supply air temperature, fan speed and valve position, respectively. Equation 24 is the same as 2, but with only one SO variable. For SOC with an optimal reference $h_0 = 0$, $T_{sa}^* = T_{sa}^{opt}$, $f_s^* = f_s^{opt}$, $v_p^* = v_p^{opt}$. For SOC with non-optimal reference point $T_{sa}^* = T_{sa}^r$, $f_s^* = f_s^r$, $v_p^* = v_p^r$.

The SO variable is controlled to zero by manipulating the compressor speed $k_s$, which is calculated with a PI controller as

$$k_s = PI_c(c) \tag{25}$$

$PI_c(\cdot)$ is a function that represents any discrete or continuous PI controller that drives the SO variable to its setpoint.

## V. Simulation

In this section, a simulation of the four different control structures is used to compare their performance. Step changes in the outdoor air temperature from its nominal value are entered in the system and the resulting COP values are recorded. The baseline results are obtained with the ESC formulation shown in the work of Salsbury *et al.* [10]. Before performing the simulations, the parameters $\mathbf{h}_0$ and $\mathbf{H}$ have to be identified; this is shown next.

### A. Identification of $\mathbf{h}_0$ and $\mathbf{H}$

In order to identify $\mathbf{h}_0$ and $\mathbf{H}$, step tests of process inputs and disturbances are performed, the data are recorded, and the procedures described in the Appendices A, B and C are implemented.

The step tests are based on the nominal conditions of the outdoor air temperature and zone load, which are $T_{oa}^* = 27$ °C and $Q^* = 2900$ W. The setpoints for the zone temperature and superheat temperature are $r_{zn} = 24$ °C and $r_{sh} = 6$ °C.

The nominal optimal operating conditions for these disturbance values are obtained with ESC; the optimal value for the compressor speed is $k_s^{opt} = 0.4955$, supply air temperature $T_{sa}^{opt} = 10.1674$ °C, fan speed $f_s^{opt} = 0.5457$ and valve position $v_p^{opt} = 0.1233$. It should be noted that the values for $k_s$, $f_s$ and $v_p$ are between 0 and 1, where 0 implies minimum speed or position, and 1 implies maximum speed or position.

The non-optimal reference value selected for the compressor speed is $k_s^r = 0.4805$, supply air temperature $T_{sa}^r = 12$ °C, fan speed $f_s^r = 0.6022$ and valve position $v_p^r = 0.1279$.

The sequence of input and disturbances are generated as $Q_i = Q^* + \Delta Q_i$, $T_{oa,j} = T^*_{oa} + \Delta T_{oa,j}$ and $k_{s,k} = k^*_s + \Delta k_{s,k}$ where $k^*_s = k^{opt}_s$ for SOC, and $k^*_s = k^r_s$ for SOC-NOR; furthermore,

$$\Delta Q_i = -120 + 20i, \quad i \in [1, 2, ..., n_Q] \tag{26}$$

$$\Delta T_{oa,j} = -6 + j, \quad j \in [1, 2, ..., n_T] \tag{27}$$

$$\Delta k_{s,k} = -0.43 + 0.08k, \quad k \in [1, 2, ..., n_k] \tag{28}$$

with $n_Q = 11$, $n_T = 16$ and $n_k = 11$ being the number of values for the zone load, outdoor air temperature and compressor speed, respectively. This gives us 1936 tests with all the possible combinations of $Q_i$, $T_{oa,j}$ and $k_{s,k}$. It is shown in Appendix C that the number of samples need to identify $\mathbf{h}_0$ and $\mathbf{H}$ is at least $\frac{n_y^2 + 3n_y}{2}$; since $n_y = 3$, we have more samples than the 9 needed. After performing the step tests, the calculated value of $\mathbf{H}$ for SOC is

$$\mathbf{H} = \begin{bmatrix} 0.3969017 & -124.6954 & 216.0882 \end{bmatrix} \tag{29}$$

and the parameters for SOC-NOR are

$$\mathbf{h}_0 = -3.8704 \tag{30}$$

$$\mathbf{H} = \begin{bmatrix} 0.2496113 & -74.37981 & 142.5401 \end{bmatrix} \tag{31}$$

### B. Performance Evaluation

The outdoor air temperature is changed from its nominal value in order to evaluate the performance of the control structures. The number of $T_{oa}$ values used is 14 and are selected as $T_{oa,l} = T^*_{oa} + \Delta T_{oa,l}$, where

$$\Delta T_{oa,l} = -4 + l, \quad l \in [1, 2, ..., 14] \tag{32}$$

The rest of the nominal operating conditions do not change. The simulations run for 70000 seconds. The system is initially driven to its nominal optimal conditions, and changes in the outdoor air temperature are introduced at the 50000 second mark.

### C. Results

Figure 3 shows how the COP changes when the outdoor air temperature changes. The line colors and types correspond to the control structures; red solid for standard control (Std), green short dashed for ESC, blue medium dashed for SOC, and purple long dashed for SOC-NOR. As expected, the COP decreases from its nominal value when the $\Delta T_{oa} > 0$, and increases when $\Delta T_{oa} < 0$. We can see that the COP results for ESC, SOC and SOC-NOR are essentially identical for most of the temperature range, whereas the COP obtained with standard control is lower than that of ESC when $\Delta T_{oa}$ increases. We assume that the ESC results correspond to the optimal operation and conclude that SOC and SOC-NOR kept the VAV system at its optimal operation for the simulated disturbance conditions. However, SOC-NOR used normal non-optimal operation data to identify its parameters, while SOC needed optimal data for the same purpose.

An interesting result is that COP values obtained with ESC, SOC and SOC-NOR start diverging when $\Delta T_{oa} > 8$ °C. We need to explore why this happens, but such exploration is out of the scope of this work.

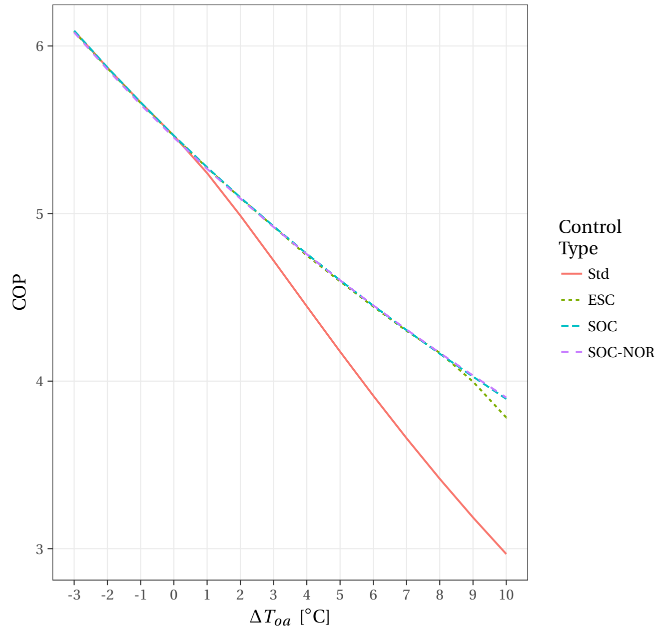



Fig. 3: COP for the different control structures

## VI. Conclusions

We have developed a method to perform SOC without the need to have optimal data as reference points to calculate the SO variable. The obtained results are very similar to the ones obtained with the SOC approach that requires optimal data, and also with ESC which was used as a benchmark for optimal performance.

## Appendix

### A. Derivation of $\mathbf{H}$ and $\mathbf{h}_0$

In order to calculate $\mathbf{H}$ explicitly from measured data, we substitute Equation 12 into Equation 6 and express $\Delta J$ as

$$\Delta J = \begin{bmatrix} \mathbf{J}^*_u \\ \mathbf{J}^*_d \end{bmatrix}^T \mathbf{G}^+_y \Delta \mathbf{y} + \frac{1}{2} \Delta \mathbf{y}^T (\mathbf{G}^+_y)^T \begin{bmatrix} \mathbf{J}^*_{uu} & \mathbf{J}^*_{ud} \\ \mathbf{J}^*_{du} & \mathbf{J}^*_{dd} \end{bmatrix} \mathbf{G}^+_y \Delta \mathbf{y} \tag{33}$$

Defining

$$\mathbf{J}_y = \left(\mathbf{G}^+_y\right)^T \begin{bmatrix} \mathbf{J}^*_u \\ \mathbf{J}^*_d \end{bmatrix} \tag{34}$$

and

$$\mathbf{J}_{yy} = (\mathbf{G}^+_y)^T \begin{bmatrix} \mathbf{J}^*_{uu} & \mathbf{J}^*_{ud} \\ \mathbf{J}^*_{du} & \mathbf{J}^*_{dd} \end{bmatrix} \mathbf{G}^+_y \tag{35}$$

we simplify $\Delta J$ as

$$\Delta J = \mathbf{J}^T_y \Delta \mathbf{y} + \frac{1}{2} \Delta \mathbf{y}^T \mathbf{J}_{yy} \Delta \mathbf{y} \tag{36}$$

Looking into $\mathbf{J}_{yy}$ and Equation 15 we see that

$$\mathbf{J}_{yy} = (\mathbf{G}^+_y)^T \begin{bmatrix} \begin{bmatrix} \mathbf{J}^*_{uu} & \mathbf{J}^*_{ud} \end{bmatrix} \mathbf{G}^+_y \\ \begin{bmatrix} \mathbf{J}^*_{du} & \mathbf{J}^*_{dd} \end{bmatrix} \mathbf{G}^+_y \end{bmatrix} = (\mathbf{G}^+_y)^T \begin{bmatrix} \mathbf{H} \\ \begin{bmatrix} \mathbf{J}^*_{du} & \mathbf{J}^*_{dd} \end{bmatrix} \mathbf{G}^+_y \end{bmatrix} \tag{37}$$

Premultiplying by $\mathbf{G}_y^T$, and using Equation 11, we get

$$\begin{bmatrix} \mathbf{H} \\ [\mathbf{J}_{du}^* \quad \mathbf{J}_{dd}^*]\,\mathbf{G}_y^+ \end{bmatrix} = \mathbf{G}_y^T \mathbf{J}_{yy} = \begin{bmatrix} (\mathbf{G}_u^*)^T \\ (\mathbf{G}_d^*)^T \end{bmatrix} \mathbf{J}_{yy} = \begin{bmatrix} (\mathbf{G}_u^*)^T \mathbf{J}_{yy} \\ (\mathbf{G}_d^*)^T \mathbf{J}_{yy} \end{bmatrix} \quad (38)$$

Therefore

$$\mathbf{H} = (\mathbf{G}_u^*)^T \mathbf{J}_{yy} \quad (39)$$

Now we have to calculate $\mathbf{J}_u^*$. If we premultiply Equation 34 by $\mathbf{G}_y^T$, and substitute this value with Equation 11 we get

$$\begin{bmatrix} \mathbf{J}_u^* \\ \mathbf{J}_d^* \end{bmatrix} = \left(\mathbf{G}_y\right)^T \mathbf{J}_y = \begin{bmatrix} (\mathbf{G}_u^*)^T \\ (\mathbf{G}_d^*)^T \end{bmatrix} \mathbf{J}_y = \begin{bmatrix} (\mathbf{G}_u^*)^T \mathbf{J}_y \\ (\mathbf{G}_d^*)^T \mathbf{J}_y \end{bmatrix} \quad (40)$$

The upper element in the last term of the previous equation is equivalent to $\mathbf{J}_u^*$, which according to Equation 14 is the same as $\mathbf{h}_0$; therefore,

$$\mathbf{h}_0 = \left(\mathbf{G}_u^*\right)^T \mathbf{J}_y \quad (41)$$

The procedure to calculate of $\mathbf{G}_u^*$, $\mathbf{J}_y$ and $\mathbf{J}_{yy}$ from measured data is shown in Appendices B and C.

### B. *Calculation of* $\mathbf{G}_u^*$

In order to calculate $\mathbf{G}_u^*$ we have to do step tests of the inputs $\mathbf{u}$ while keeping the disturbances constant at their nominal value $\mathbf{d}^*$. Each input $u_i$ in $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \dots & u_i & \dots & u_{n_u} \end{bmatrix}^T$ is stepped through several values around the nominal point $u_i^*$, while the other inputs $u_j$, $j \neq i$, are kept constant at their nominal values $u_j^*$; thus, after each step change in $u_i$, we wait until the plant reaches a steady state $y_i$, then we record all the values in $\mathbf{y} = \begin{bmatrix} y_1 & y_2 & \dots & y_i & \dots & y_{n_y} \end{bmatrix}^T$.

For each step test $k$, we record the difference $\Delta y_{i,k} = y_{i,k} - y_{i,k}^*$ and $\Delta u_{j,k} = u_{j,k} - u_{j,k}^*$, and form the vectors $\Delta\mathbf{y}_k = [\Delta y_{1,k} \quad \Delta y_{2,k} \quad \dots \quad \Delta y_{n_y,k}]^T$ and $\Delta\mathbf{u}_k = [\Delta u_{1,k} \quad \Delta u_{2,k} \quad \dots \quad \Delta u_{n_u,k}]^T$. From Equation 8 we see that $\Delta\mathbf{y}_k = \mathbf{G}_u^* \Delta\mathbf{u}_k$. After $K$ step tests we can gather the measurements as

$$\begin{bmatrix} \Delta\mathbf{y}_1 & \Delta\mathbf{y}_2 & \cdots & \Delta\mathbf{y}_K \end{bmatrix} = \mathbf{G}_u^* \begin{bmatrix} \Delta\mathbf{u}_1 & \Delta\mathbf{u}_2 & \cdots & \Delta\mathbf{u}_K \end{bmatrix} \quad (42)$$

and make $\Delta\mathbf{Y} = [\Delta\mathbf{y}_1 \quad \Delta\mathbf{y}_2 \quad \cdots \quad \Delta\mathbf{y}_K]$ and $\Delta\mathbf{U} = [\Delta\mathbf{u}_1 \quad \Delta\mathbf{u}_2 \quad \cdots \quad \Delta\mathbf{u}_K]$; then, we solve for $\mathbf{G}_u^*$ as

$$\mathbf{G}_u^* = \Delta\mathbf{Y}\Delta\mathbf{U}^T \left(\Delta\mathbf{U}\Delta\mathbf{U}^T\right)^{-1} \quad (43)$$

### C. *Calculation of* $\mathbf{J}_y$ *and* $\mathbf{J}_{yy}$

Similar to the previous section, the vector $\Delta\mathbf{y}$ is formed with the differences, $\Delta y_i = y_i - y_i^*$, between the individual measurements and their nominal values as

$$\Delta\mathbf{y} = \begin{bmatrix} \Delta y_1 & \Delta y_2 & \dots & \Delta y_{n_y} \end{bmatrix}^T \quad (44)$$

We can expand Equation 36 to express $\Delta J$ as a sum of the individual $\Delta y_i$ as

$$\begin{aligned} \Delta J = & a_1 \Delta y_1 + \cdots + a_{n_y} \Delta y_{n_y} + a_{n_y+1} \Delta y_1^2 + a_{n_y+2} \Delta y_1 \Delta y_2 + \cdots \\ & \cdots + a_{\frac{n_y^2+3n_y}{2}-1} \Delta y_{n_y-1} \Delta y_{n_y} + a_{\frac{n_y^2+3n_y}{2}} \Delta y_{n_y}^2 \end{aligned} \quad (45)$$

This means that the matrices $\Delta\mathbf{J}_y$ and $\Delta\mathbf{J}_{yy}$ can be represented in terms of the coefficients $a_i$ as

$$\mathbf{J}_y = \begin{bmatrix} a_1 & a_2 & \cdots & a_{n_y} \end{bmatrix}^T \quad (46)$$

$$\mathbf{J}_{yy} = \begin{bmatrix} 2a_{n_y+1} & a_{n_y+2} & \cdots & a_{2n_y} \\ a_{n_y+2} & 2a_{2n_y+1} & \cdots & a_{3n_y-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{2n_y} & a_{3n_y-1} & \cdots & 2a_{\frac{n_y^2+3n_y}{2}} \end{bmatrix} \quad (47)$$

Going back to Equation 45, we can simplify the expression of $\Delta J$ as

$$\Delta J = \mathbf{z}^T \mathbf{a} \quad (48)$$

where

$$\mathbf{z} = \begin{bmatrix} \Delta y_1 & \cdots & \Delta y_n & \Delta y_1^2 & \Delta y_1 \Delta y_2 & \cdots & \Delta y_{n_y}^2 \end{bmatrix}^T \quad (49)$$

and

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & \cdots & a_{\frac{n_y^2+3n_y}{2}} \end{bmatrix}^T \quad (50)$$

For each input step in $u_i$ we would get a pair of values $\Delta J_i$ and $\mathbf{z}_i$. With $M$ step tests, we would have the following data set

$$\begin{bmatrix} \Delta J_1 & \Delta J_2 & \vdots & \Delta J_M \end{bmatrix}^T = \begin{bmatrix} \mathbf{z}_1 & \mathbf{z}_2 & \vdots & \mathbf{z}_M \end{bmatrix}^T \mathbf{a} \quad (51)$$

Naming $\mathbf{V} = \begin{bmatrix} \Delta J_1 & \Delta J_2 & \cdots & \Delta J_M \end{bmatrix}^T$ and $\mathbf{Z} = \begin{bmatrix} \mathbf{z}_1 & \mathbf{z}_2 & \cdots & \mathbf{z}_M \end{bmatrix}^T$, we can solve for the vector of matrix elements $\mathbf{a}$ as

$$\mathbf{a} = \left(\mathbf{Z}^T \mathbf{Z}\right)^{-1} \mathbf{Z}^T \mathbf{V} \quad (52)$$

With this vector we can now form the matrices $\mathbf{J}_y$ and $\mathbf{J}_{yy}$ using Equations 46 and 47. It should be noted that the number of samples $M$ should be greater than, or equal to, $\frac{n_y^2+3n_y}{2}$ in order to have a full column rank matrix $\mathbf{Z}$, so that $\mathbf{Z}^T\mathbf{Z}$ is invertible.

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